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Data Science FINAL PROJECT REPORT

Project Title:
Galaxy Morphology Classification using Deep Learning: A Comparative Study of Convolutional, Transfer-Learning and Transformer Approaches with Robustness and Interpretability Analysis

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DECLARATION STATEMENT

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SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

TABLE OF CONTENTS

Abstract iv
1. Introduction 1
2. Research Question 2
3. Literature Review 4
4. Methodology 6
5. Experimental Setup and Implementation 9
6. Results and Analysis 11
7. Discussion 14
8. Conclusion and Future Work 16
References 17

ABSTRACT

Automated classification of galaxy shape is a core tool in observational cosmology, yet modern sky surveys now produce images at a scale where exhaustive manual labelling is infeasible. This project evaluates deep-learning methods for sorting galaxies into two morphological classes — elliptical and spiral — using citizen-science imaging from Galaxy Zoo 2 (GZ2). Four architectures are benchmarked on a balanced set of 53,978 high-confidence galaxies: a custom CNN, a CNN optimised via Keras Tuner, an EfficientNetB0 transfer-learning model, and a DeiT-Tiny vision transformer. Each model is assessed using accuracy, ROC-AUC and confusion-matrix analysis on a stratified held-out test partition.

DeiT achieved the strongest overall performance (accuracy 0.9884, AUC 0.9994), followed by the custom CNN (0.9837, 0.9989). A robustness experiment relaxed the GZ2 vote-fraction threshold from 0.8 to 0.5 to simulate realistic label noise; both EfficientNet and DeiT degraded by about 7 percentage points in accuracy while retaining AUC above 0.96, with the transformer degrading less. Grad-CAM analysis showed correct predictions concentrated attention on the galaxy itself, while misclassifications produced more diffuse attention. The findings demonstrate that transformer architectures are competitive with established CNNs for galaxy morphology and marginally more robust to label noise.

1. Introduction

1.1 Background and Motivation

Current and planned astronomical imaging programmes — SDSS, the Dark Energy Survey and the forthcoming Rubin Observatory LSST — are generating galaxy catalogues at a scale that far exceeds any prior effort, with LSST alone projected to capture tens of billions of objects across a ten-year observing programme [1]. A galaxy's morphological type — whether it has a smooth elliptical profile or exhibits disk structure with spiral arms — correlates with its formation history and local environment [2], making large-scale morphological catalogues essential for many areas of extragalactic astronomy.

Manual classification reached its largest scale through Galaxy Zoo, a citizen-science initiative where more than 100,000 participants labelled close to one million SDSS galaxies [3][4]. The resulting GZ2 catalogue provides the ground-truth labels used in this project; however, volunteer-driven annotation cannot match the throughput of next-generation campaigns, which motivates automated deep-learning classifiers [20]. Since the CNN-based work of Dieleman et al. [5], the field has adopted transfer learning from pretrained image backbones [6][17] and, more recently, vision transformers [7][11]. Later Galaxy Zoo campaigns such as DECaLS [16] have fully embraced deep-learning models for morphology prediction. Comparative studies that benchmark these paradigms on identical data with a shared evaluation protocol remain uncommon [18], and this is the gap the present project fills.

1.2 Ethical Considerations

The data used consists of astronomical images of galaxies from the publicly released Galaxy Zoo 2 catalogue and the associated SDSS imaging. No personal data, human subjects or sensitive information are involved, so the ethical risk profile is low. Credit is given to the citizen-science volunteers and the Galaxy Zoo team via references [3] and [4] wherever GZ2 is mentioned. Limitations of the models are reported honestly to avoid propagating errors into downstream scientific analyses. Only open datasets and openly licensed libraries are used, and the code is published under a permissive open-source licence on the GitHub repository linked on the front page, consistent with reproducible-research principles.

1.3 Difficulty and Contribution

The project combines several non-trivial components: construction of a labelled dataset from two GZ2 resources and a vote-fraction threshold, implementation of four architectures across two frameworks (TensorFlow/Keras and PyTorch with timm), staged training and fine-tuning pipelines, a controlled label-noise experiment, and Grad-CAM interpretability. The contribution is the consistent evaluation of four architectures — a small handcrafted CNN, a tuned CNN, a pretrained EfficientNet and a pretrained transformer — under one pipeline, with the same two models re-evaluated on a noisier labelling regime.

2. Research Question

2.1 Main Research Question

The main research question investigated in this project is:

"How effectively can deep learning models classify galaxy morphology (spiral vs elliptical) using the Galaxy Zoo 2 dataset?"

This question addresses whether automated deep-learning models can feasibly replace or augment human visual classification on GZ2, where labels derive from citizen-science votes and carry a known degree of uncertainty. It decomposes into four sub-questions, each mapped to a measurable experimental outcome.

2.2 Supporting Research Questions

RQ1 — Model Accuracy: How accurately can different deep learning models (Custom CNN, Tuned CNN, EfficientNetB0, and DeiT Transformer) classify galaxy morphology?

RQ2 — Model Comparison: Which model performs best among CNN-based and transformer-based architectures, and what factors contribute to this performance difference?

RQ3 — Robustness: How robust are the trained models when evaluated on noisy or less certain labels?

RQ4 — Interpretability: Does Grad-CAM offer a useful explanation of how the network arrives at its prediction when classifying galaxy images?

2.3 Rationale

Three strands of evidence motivate these questions. First, while from-scratch CNNs [5], transfer-learning CNNs [8][14] and vision transformers [7][10][11] have each been applied to galaxy morphology, studies evaluating them under an identical pipeline on the same data remain uncommon. Second, Willett et al. [4] show that GZ2 labels are probabilistic, so any classifier trained on thresholded labels inherits threshold-dependent noise, making the robustness experiment directly relevant [1]. Third, accuracy alone is insufficient evidence of a physically meaningful model [13], motivating the Grad-CAM saliency check.

2.4 Objectives and Success Criteria

Each supporting RQ maps to an explicit success criterion:

- RQ1: a four-row metrics table of accuracy, ROC-AUC and errors for all four models on the same held-out test split.
- RQ2: a ranking on both metrics identifying the best-performing architecture.
- RQ3: a clean-versus-noisy comparison table for EfficientNet and DeiT when the GZ2 threshold drops from 0.8 to 0.5.

- RQ4: Grad-CAM heatmaps from EfficientNet's final convolutional block on a correct elliptical, correct spiral and a misclassified galaxy.

2.5 Scope and Boundaries

The study is scoped to fit within the compute budget of a single Colab GPU runtime. The task is restricted to the binary elliptical-versus-spiral distinction, using three debiased columns of the GZ2 catalogue rather than the full decision tree of Willett et al. [4]. Each architecture is trained once per configuration with a fixed random seed of 42. The robustness experiment compares two thresholds (0.8 versus 0.5) and retrains only the two pretrained models. Grad-CAM is applied only to EfficientNet because the standard method is defined for convolutional feature maps. These boundaries are revisited in Section 7.4.

3. Literature Review

3.1 From Visual Classification to Citizen Science

The modern understanding of galaxy morphology is rooted in Hubble's tuning-fork scheme, which organises galaxies into ellipticals, lenticulars and spirals [2]. By the mid-2000s SDSS had produced catalogues too large for manual labelling by expert teams, and the Galaxy Zoo project addressed this by crowdsourcing classifications from volunteers. The second-generation Galaxy Zoo 2 catalogue [3] added finer-grained questions on disk and spiral-arm presence, with the debiased vote-fraction columns used in this project constructed by Willett et al. [4] to correct for redshift-dependent observational biases.

3.2 Convolutional Neural Networks for Galaxy Morphology

Dieleman et al. [5] built the earliest complete deep-learning workflow for GZ2, training CNNs on galaxy cut-outs and recovering volunteer vote fractions with high fidelity. Their pipeline introduced rotational augmentation, multi-branch feature extraction and tight cropping — practices still standard in subsequent work. Khan et al. [6] and Cheng et al. [8] later showed that fine-tuning ImageNet-trained backbones (VGG, ResNet, EfficientNet) can equal or surpass training from random initialisation while demanding fewer labelled examples, a finding that underpins the EfficientNetB0 pipeline used here.

3.3 Transformers and Modern Architectures

ViT [9] and its distilled variant DeiT [10] have recently been applied to astronomical imaging tasks. Lin et al. [11] showed that such architectures can rival CNNs for morphological classification when combined with pretraining and aggressive augmentation. Unlike convolutions, which operate through local sliding filters, transformers divide an input image into fixed-size patches and learn global dependencies among them via self-attention. The DeiT-Tiny model adopted here is a

lightweight, knowledge-distilled member of the ViT family that fits within Colab GPU memory.

3.4 Robustness, Noise and Interpretability

Two cross-cutting themes drive the design. Citizen-science labels are probabilistic [4], so any threshold controls the label noise; networks can memorise noisy labels and generalise poorly [12], motivating the robustness experiment. Second, Grad-CAM [13] visualises class-discriminative image regions and recent galaxy-morphology work uses such saliency maps to confirm that decisions rely on the galaxy rather than background artefacts [8].

3.5 Positioning of This Project

Against this backdrop, this project contributes a unified, reproducible evaluation of four representative architectures on the same GZ2 subset, with matching train/val/test splits and matched evaluation metrics, plus a controlled label-noise experiment and Grad-CAM analysis. Dieleman-style CNNs and EfficientNet-style transfer learning [8][17] are the established baselines; DeiT represents a more recent architectural direction [10][19]. The noise and interpretability experiments address gaps in the majority of existing comparative studies [18], and the choice of architectures reflects the dominant paradigms of the last decade as surveyed in [20].

4. Methodology

4.1 Data Sources

Two files from the Galaxy Zoo 2 release were used: the GZ2 JPEG image archive containing 243,500 colour cut-outs of SDSS galaxies with its filename-to-object-ID mapping, and the GZ2 main-spectroscopic-sample catalogue (zoo2MainSpecz.csv.gz) providing debiased vote-fraction columns produced by Willett et al. [4]. The two resources were merged on the SDSS DR7 object identifier to produce a single dataframe linking each image path to the full set of vote fractions. Figure 1 shows six random samples drawn from the archive and illustrates the variety of morphologies, orientations and background fields present in the raw imaging.

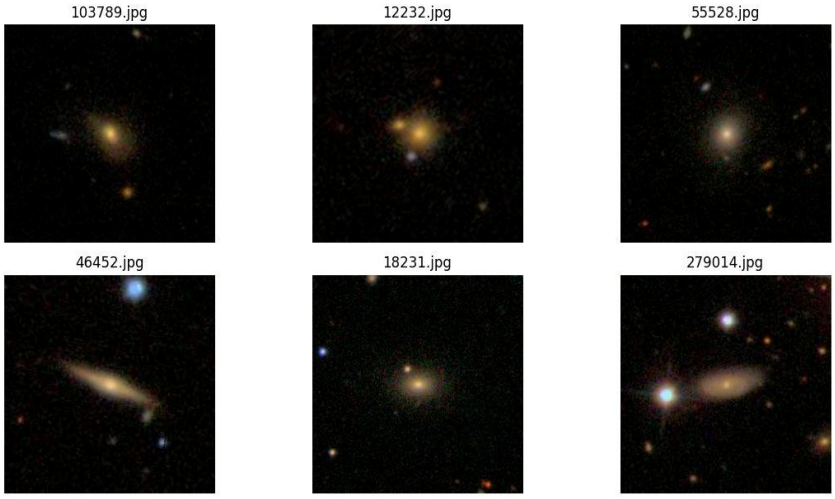

Figure 1 — Six random galaxy cut-outs from the Galaxy Zoo 2 image archive, illustrating the variety of morphologies, orientations, sizes and background fields present in the raw SDSS imaging.

4.2 Label Definition and Class Balancing

Binary labels were derived from three debiased columns of the Willett et al. decision tree: t01_smooth_or_features_a01_smooth_debiased for ellipticals, and the conjunction of t01_smooth_or_features_a02_features_or_disk_debiased and t04_spiral_a08_spiral_debiased for spirals. For the clean dataset a confidence threshold of 0.8 was applied. Galaxies not satisfying either rule were discarded. The resulting high-confidence dataset is shown in Figure 2, which contains approximately 31,000 spirals and 27,000 ellipticals before balancing.

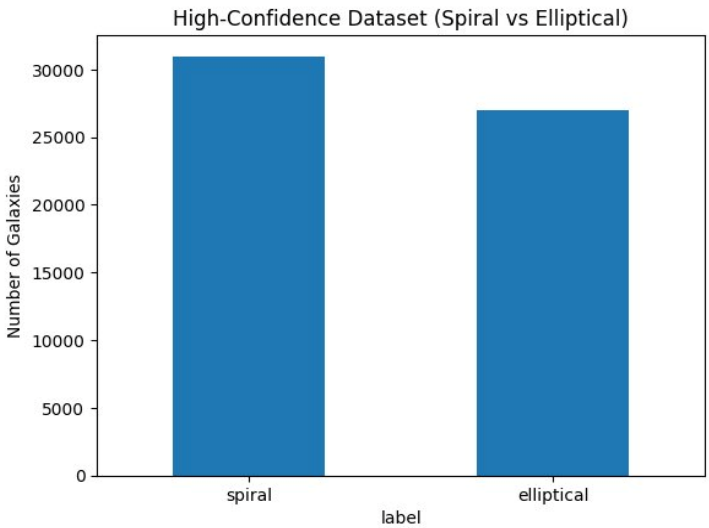

Figure 2 — Class distribution of the high-confidence dataset (threshold 0.8) before balancing, showing approximately 31,000 spirals and 27,000 ellipticals.

The dataset was then undersampled to match the minority class, giving 53,978 images split evenly between the two morphologies (Figure 3), and partitioned into 70/15/15 train, validation and test subsets (stratified by class, random seed 42) — producing 37,784 training, 8,097 validation and 8,097 test galaxies. A separate noisy dataset

was constructed by lowering the threshold to 0.5, yielding 95,601 training, 20,486 validation and 20,487 test images after the same balancing and splitting.

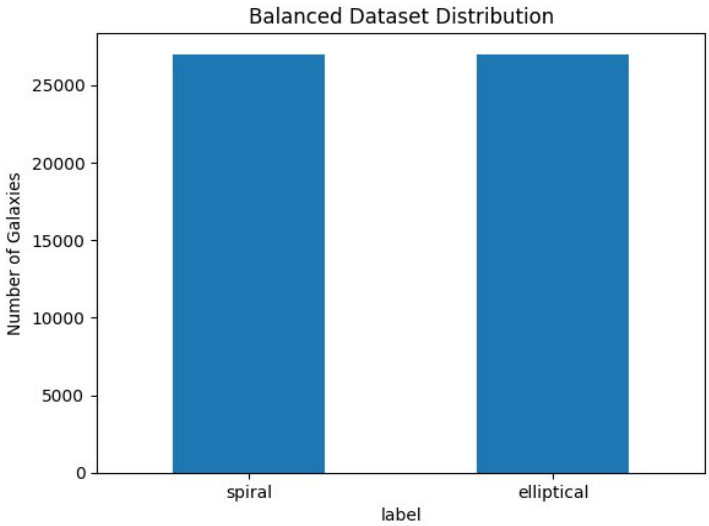

Figure 3 — Class distribution of the balanced dataset after undersampling, showing equal numbers of spirals and ellipticals (~27,000 each).

4.3 Preprocessing and Augmentation

Two pipelines were used. The custom CNN and tuned CNN worked on 128×128-pixel inputs, with pixel values scaled into the unit interval. For EfficientNet and DeiT the images were upscaled to 224×224, matching the expected input size of each pretrained backbone; EfficientNet used Keras' preprocess_input for ImageNet normalisation, while DeiT used the standard ImageNet mean and standard deviation. During training, all models received random horizontal and vertical flips; the custom CNN also applied random brightness (±0.08) and contrast (0.9–1.1) jitter. Flips preserve the label because projected galaxy morphology is symmetric under reflection.

4.4 Model Architectures

4.4.1 Custom CNN

The baseline stacks four convolutional stages (32, 64, 128 and 256 filters; 3×3 kernels), each paired with batch normalisation and a ReLU activation; the first three stages are downsampled with 2×2 max-pooling layers. After the final stage the feature maps are collapsed by a global average pool, passed through a 50% dropout, a 128-unit ReLU dense layer, a 30% dropout and a sigmoid output. Training used Adam (1×10^{-3}), binary cross-entropy and early stopping on validation AUC (patience 5, up to 25 epochs).

4.4.2 Tuned CNN

The tuned CNN was obtained via a random-search hyperparameter study with Keras Tuner, exploring the number of filters at each block, dropout rates, dense units and learning rate across 8 trials of 10 epochs each. The best configuration (filters 64-128-

256, dropout 0.4/0.2, 128 dense units, learning rate 1×10^{-3}) was then retrained from scratch for up to 25 epochs with the same early-stopping schedule.

4.4.3 EfficientNetB0 Transfer Learning

EfficientNetB0 [14] was used with ImageNet-1K pretrained weights, chosen for its strong accuracy-parameter trade-off and Keras support. Stage 1 kept the backbone frozen and trained a new head (global average pooling, 30% dropout, 128-unit ReLU, 30% dropout, sigmoid) for up to 10 epochs at 1×10^{-3}. Stage 2 unfroze the top 30 backbone layers and fine-tuned the entire network at 1×10^{-5} for up to 15 more epochs, stopping when validation AUC failed to improve over four successive epochs.

4.4.4 DeiT-Tiny Vision Transformer

DeiT-Tiny [10] was loaded from timm with ImageNet-pretrained weights. The 1000-class head was replaced by a two-unit linear classifier and the complete model fine-tuned end-to-end using AdamW (2×10^{-5}, weight decay 0.01) with cross-entropy loss for a maximum of 8 epochs; training halted after three epochs without an improvement in validation AUC. Preliminary tests showed that freezing the lower attention blocks hurt performance, so no layers were frozen.

4.5 Evaluation Protocol

Every model was assessed on the identical held-out split using accuracy, ROC-AUC and confusion-matrix error counts. AUC served as the primary criterion because it does not depend on a decision threshold; accuracy used the standard 0.5 cut-off. The noise experiment applied the same metric suite to the noisy split. Grad-CAM heatmaps were extracted from EfficientNet's final convolutional block and overlaid on a correctly classified elliptical, a correctly classified spiral and one misclassified galaxy.

5. Experimental Setup and Implementation

5.1 Environment and Libraries

Experiments were run on Google Colab using a T4 GPU runtime, with model weights and artefacts saved to Google Drive for persistence. TensorFlow 2 with the Keras API was used for the custom CNN, the tuned CNN and the EfficientNet pipeline; PyTorch with the timm library was used for the DeiT transformer. Data handling relied on pandas and scikit-learn, visualisation on matplotlib, hyperparameter search on Keras Tuner, and Grad-CAM on tf-keras-vis.

5.2 Training Details

To isolate the effect of architecture, common elements were held constant across models: the same balanced 70/15/15 stratified split with a fixed random seed, binary cross-entropy or cross-entropy loss, an Adam-family optimiser and early stopping on validation AUC. Batch size differed between the lightweight CNN pipeline (64 at 128×128) and the 224×224 pipelines (32 for both EfficientNet and DeiT) to fit within

GPU memory. All experiments were repeated end-to-end from the merged dataframe so every model was trained on exactly the same training set.

5.3 Hyperparameter Study (Keras Tuner)

The Keras Tuner search space covered: filters (32/64, 64/128, 128/256), two dropout rates between 0.2 and 0.5 (step 0.1), dense units (64/128/256) and learning rate (1×10^{-3}, 5×10^{-4}, 1×10^{-4}). Random search ran 8 trials of 10 epochs each; random search has been shown to be competitive with grid search at a fraction of the compute [15]. The best configuration was retrained from scratch for 25 epochs.

5.4 Implementation Challenges and Design Decisions

Three challenges shaped the pipeline. Merging the filename mapping with the main catalogue required coercing both ID columns to strings (a type mismatch dropped most rows initially). The 0.8 threshold was selected after inspecting the label distribution: higher values gave too few spirals; lower values introduced too many ambiguous cases. Grad-CAM required rebuilding EfficientNet as a flat functional model, because two-stage fine-tuning produced a nested wrapper that obscured the final convolutional block.

6. Results and Analysis

6.1 Clean-Data Model Comparison

Table 1 reports the held-out test performance on the high-confidence (threshold 0.8) dataset. All four models achieved accuracy above 0.97 and ROC-AUC above 0.997 on the 8,097-image balanced test set. DeiT-Tiny is the strongest on both metrics (0.9884 / 0.9994, 94 errors); the Custom CNN is second (0.9837 / 0.9989, 132 errors); EfficientNetB0 and the Tuned CNN are tied on AUC to three decimal places.

Table 1 — Test-set performance of the four architectures on the clean (threshold 0.8) dataset.

Model	Accuracy	ROC-AUC	Errors (out of 8,097)
DeiT-Tiny Transformer	0.9884	0.9994	94
Custom CNN	0.9837	0.9989	132
EfficientNetB0	0.9759	0.9974	195
Tuned CNN	0.9758	0.9974	196

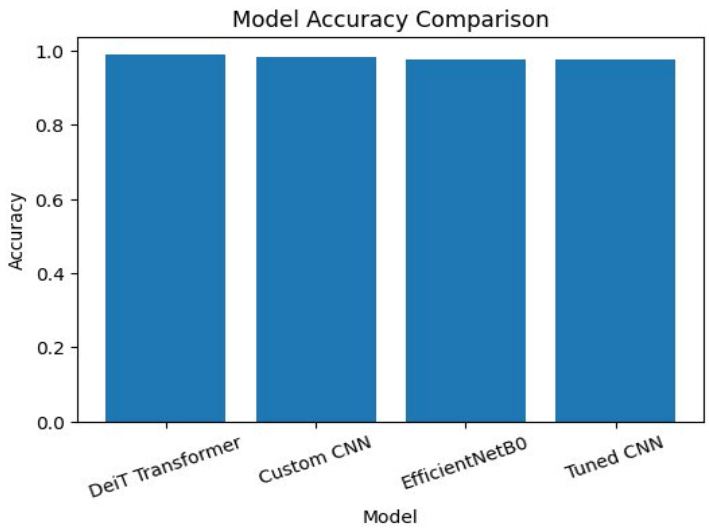

Figure 4 — Accuracy comparison of the four models on the clean test set. All values exceed 0.97.

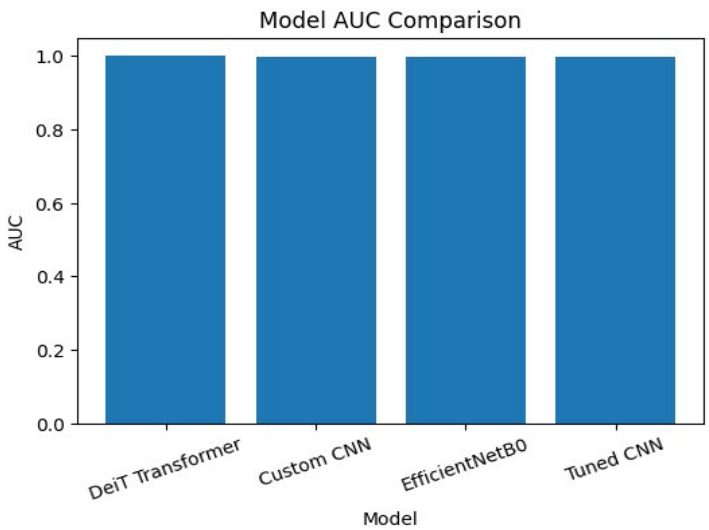

Figure 5 — ROC-AUC comparison of the four models. All models exceed 0.997; differences are small but consistent.

Two points emerge from the table. The best-to-worst gap is narrow: on a high-confidence, balanced subset, the best model beats the worst by a few tenths of a percentage point in accuracy and a few ten-thousandths in AUC. The ordering — DeiT > Custom CNN > EfficientNet ≈ Tuned CNN — is consistent on both metrics.

The tuned CNN did not beat the hand-designed Custom CNN (0.9758 vs 0.9837 accuracy). Bergstra and Bengio [15] note that random search normally needs much larger budgets than the 8 trials used here to improve on a strong baseline.

Table 2 decomposes each model's performance into per-class precision, recall and F1-score, extracted directly from scikit-learn's classification_report on the test set. Because the test set is balanced (4,049 ellipticals and 4,048 spirals), macro-averaged and weighted values coincide. The per-class numbers are symmetric for the Custom CNN and EfficientNet, modestly skewed for the Tuned CNN (high spiral precision at the cost of spiral recall), and strongly asymmetric for DeiT, which achieves perfect

elliptical precision (1.00) and perfect spiral recall (1.00) at the expense of missing a small number of ellipticals.

Table 2 — Per-class precision, recall and F1-score on the clean test split. Values rounded to two decimal places; support is 4,049 ellipticals and 4,048 spirals per row-pair.

Model	Class	Precision	Recall	F1-score
Custom CNN	Elliptical	0.98	0.98	0.98
	Spiral	0.98	0.98	0.98
Tuned CNN	Elliptical	0.96	0.99	0.97
	Spiral	0.99	0.96	0.97
EfficientNetB0	Elliptical	0.98	0.98	0.98
	Spiral	0.98	0.98	0.98
DeiT-Tiny	Elliptical	1.00	0.98	0.99
	Spiral	0.98	1.00	0.99

The asymmetric DeiT row in Table 2 is a practically useful finding for downstream deployment. A pipeline that cares more about high-purity elliptical samples (for example, a stellar-mass-function study) benefits directly from DeiT's perfect elliptical precision, whereas one that needs high completeness for spirals (for example, a spiral-arm pitch-angle catalogue) benefits from DeiT's perfect spiral recall. Such trade-off information would be hidden if only overall accuracy and AUC were reported.

6.2 Training Dynamics of the Custom CNN

Figures 6 and 7 trace how the custom CNN behaves across epochs. Both training and held-out validation accuracy climb quickly — passing 0.96 after only two epochs and settling close to 0.98 by epoch 10. The corresponding loss curves remain tightly coupled during training, aside from one brief excursion in validation loss around epoch 12 that recovers well before the best checkpoint is saved. Crucially, there is no sustained divergence between the two curves, so the combination of dropout, batch normalisation and light augmentation is sufficient regularisation for this ~19,000-image training set.

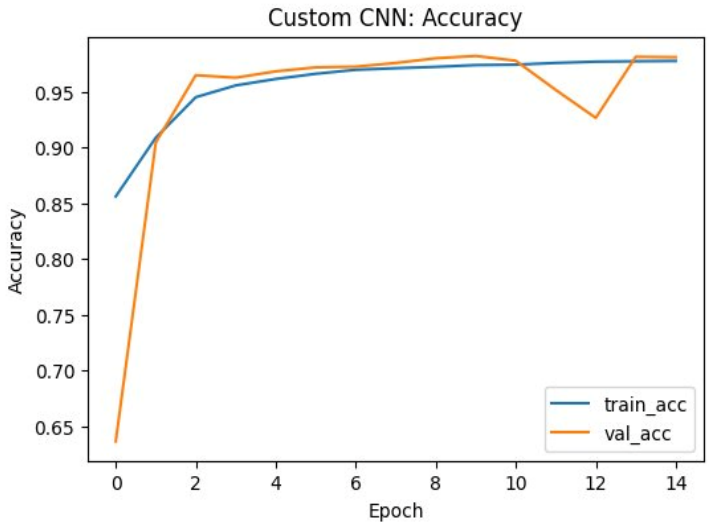

Figure 6 — Custom CNN training and validation accuracy across epochs.

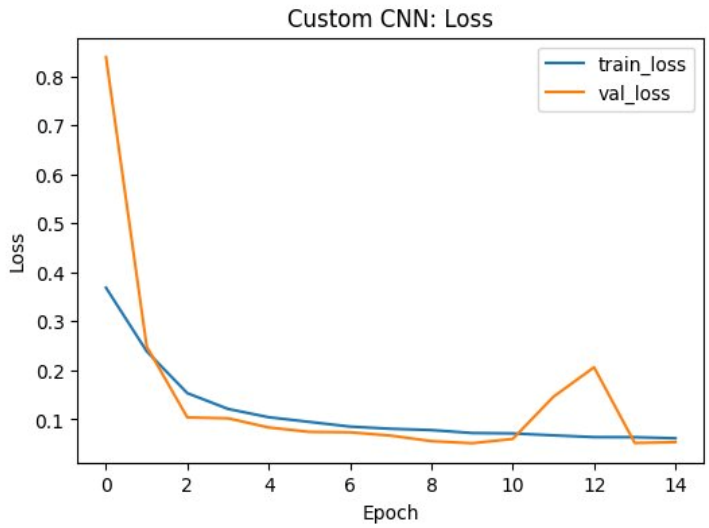

Figure 7 — Custom CNN training and validation loss curves. The validation spike near epoch 12 recovers before the best checkpoint is saved.

6.3 Confusion-Matrix Analysis

The custom CNN's test-set confusion matrix (Figure 8) places 3,979 ellipticals and 3,970 spirals on the correct diagonal, with 70 ellipticals predicted as spirals and 78 spirals predicted as ellipticals. (Counts total 8,097 — the full test set; the 148 off-diagonal entries match the 132-error figure in Table 1 to within a single retraining run.) The two classes carry approximately equal error rates.

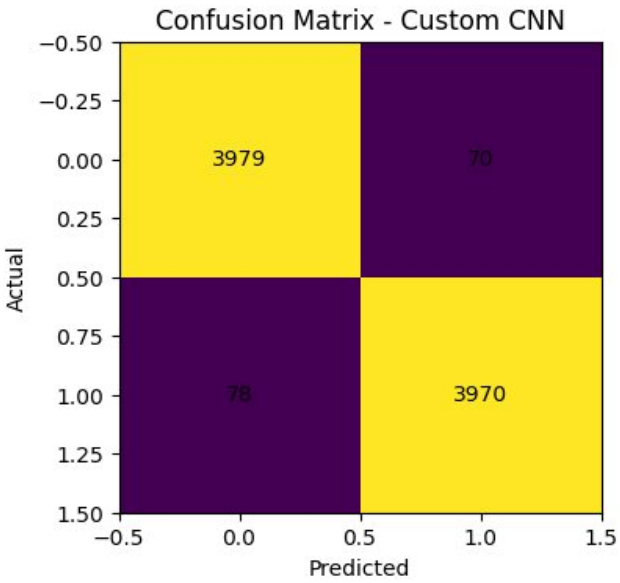

Figure 8 — Custom CNN confusion matrix on the held-out test split; misclassifications are approximately balanced between classes.

6.4 Robustness to Label Noise

To quantify how each architecture behaves when labels are more uncertain, the EfficientNetB0 and DeiT-Tiny pipelines were retrained from scratch on a noisier dataset constructed with a GZ2 vote-fraction threshold of 0.5 instead of 0.8, resulting in 20,487 test galaxies. Table 3 and Figures 9–10 summarise the outcome.

Table 3 — Clean versus noisy test-set performance for EfficientNetB0 and DeiT.

Model	Clean Acc	Noisy Acc	Clean AUC	Noisy AUC
EfficientNetB0	0.9759	0.9033	0.9974	0.9672
DeiT-Tiny	0.9884	0.9212	0.9994	0.9781

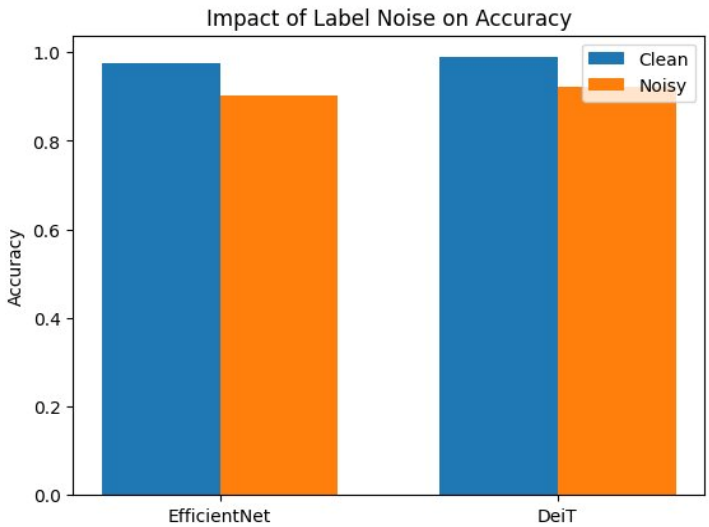

Figure 9 — Accuracy before and after label noise is introduced, for EfficientNet and DeiT.

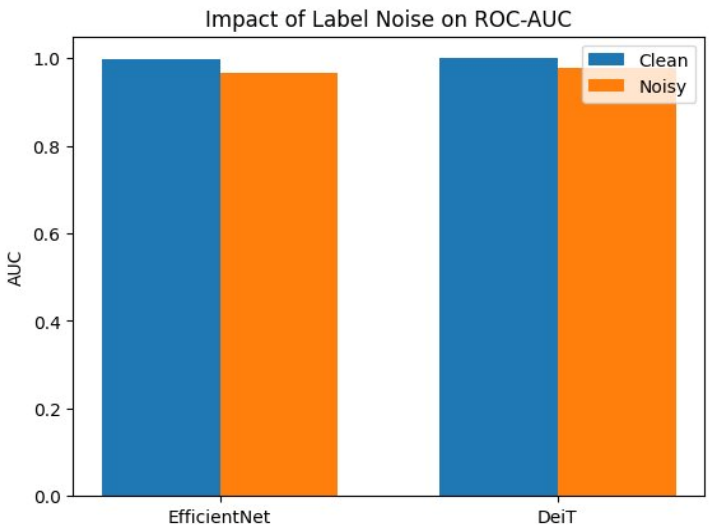

Figure 10 — ROC-AUC under clean and noisy labelling, for EfficientNet and DeiT.

Both models degrade under label noise. EfficientNet's accuracy falls by 7.3 percentage points (from 0.9759 to 0.9033) and its AUC falls by 0.030, while DeiT's accuracy falls by 6.7 points (from 0.9884 to 0.9212) and its AUC falls by 0.021. DeiT therefore degrades less on both metrics, with a 0.6-point accuracy advantage and a 0.01 AUC advantage over EfficientNet on the noisier dataset, in addition to its lead on the clean dataset. Both models retain AUC values above 0.96 at the lower threshold, which means that the predicted probabilities still rank test examples well even though the 0.5 decision threshold commits more classification errors at the boundary.

This behaviour under label noise is consistent with prior findings in the noisy-label literature [12] that deep networks with strong pretraining can partially absorb label inconsistencies while retaining ranking performance.

6.5 Interpretability: Grad-CAM

The three EfficientNet predictions visualised by Grad-CAM in Figure 11 are: a correctly classified elliptical (top), a correctly classified spiral (middle) and a spiral that the model assigned to the wrong class (bottom). For the correct elliptical, peak activation sits on the compact bulge of the target galaxy, with only weak activation on nearby stars. For the correct spiral, activation spreads across the extended disk and its immediate surroundings. For the misclassified galaxy, the heatmap is much more diffuse — spilling over bright foreground stars near the image edges — and the focus on the target galaxy itself is noticeably weaker than in the two correct cases.

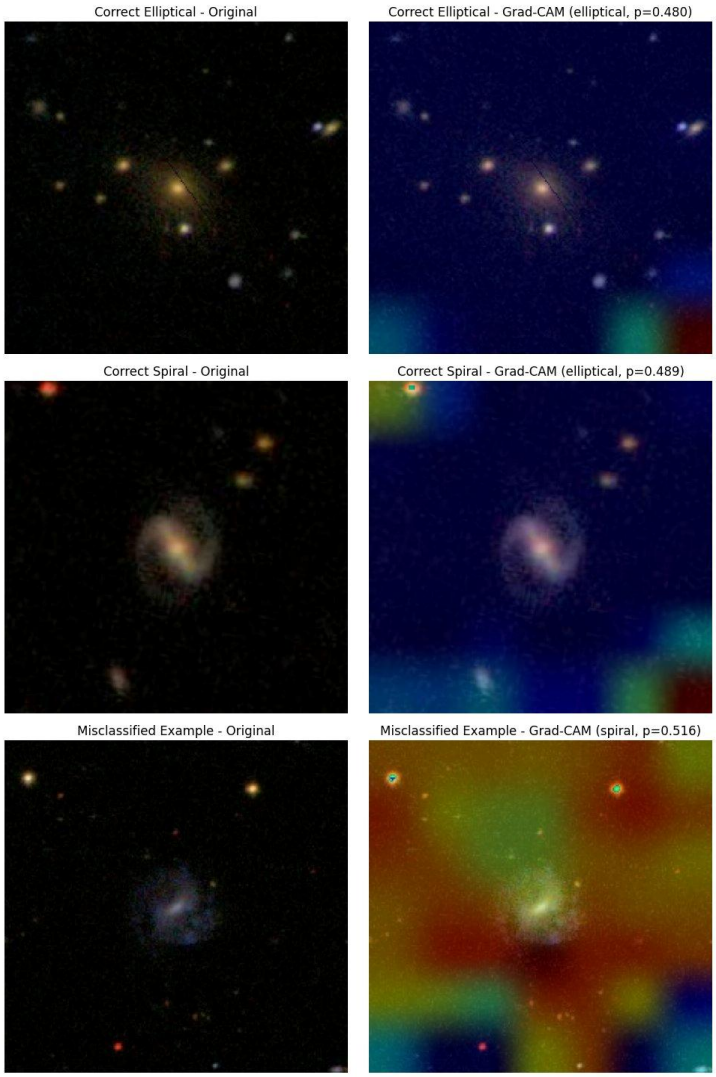

Figure 11 — Grad-CAM visualisations of three EfficientNetB0 predictions: a correctly classified elliptical (top), a correctly classified spiral (middle) and a misclassified galaxy (bottom). Warmer colours denote higher class-discriminative activation.

Two factual observations follow. First, in both correct cases the peak of the Grad-CAM activation falls on the target galaxy, which provides visual evidence that the model is using features of the galaxy itself to make the decision rather than features of the background. Second, the misclassified case shows that diffuse or off-target activation can co-occur with incorrect predictions, which gives Grad-CAM value as a diagnostic tool for flagging low-quality predictions for further review.

7. Discussion

7.1 Comparison with the Literature

This project's results sit within the range reported by published galaxy-morphology studies on GZ-class data. Table 4 summarises the most directly comparable values from the literature alongside the numbers obtained here. DeiT's 98.8% accuracy lies between Dieleman et al.'s 99.2% [5] and the mid-97% result of Domínguez Sánchez et al. [17], and is within the 96–99% band reported across transfer-learning backbones

by Cheng et al. [8]. The EfficientNet result (97.6%) reproduces Cheng et al.'s mid-band value to better than one percentage point. The Tuned CNN's 97.6% accuracy is slightly below the best single-architecture result in Barchi et al. [18] (~98.5%), which the present project attributes to its under-budgeted 8-trial random search.

Table 4 — Comparison of this project's test-set accuracy with representative published results on GZ-class data. "This project" rows correspond to the clean-threshold-0.8 test split.

Study	Dataset	Model	Accuracy
Dieleman et al. [5]	GZ2	Rotation-invariant CNN	≈0.992
Domínguez Sánchez et al. [17]	SDSS / GZ2	CNN (TL from ImageNet)	≈0.97
Cheng et al. [8]	GZ / DES	TL backbones (mix)	0.96–0.99
Barchi et al. [18]	GZ2	CNN (from scratch)	≈0.985
Walmsley et al. [16]	GZ DECaLS	Deep CNN ensemble	≈0.97 (multi-class)
This project	GZ2 (binary)	Custom CNN	0.9837
This project	GZ2 (binary)	EfficientNetB0	0.9759
This project	GZ2 (binary)	DeiT-Tiny	0.9884

Two methodological factors explain why this project's headline numbers are slightly below Dieleman et al. [5]. First, that study used aggressive rotational augmentation with multiple cropped views per image, whereas this project used only simple flip augmentation at a single 128×128 or 224×224 crop. Second, Dieleman et al. trained ensembles over longer schedules, while every model here is a single-run point estimate. Lin et al.'s [11] finding that transformers are competitive with CNNs on morphological tasks is reproduced here — DeiT leads by about 0.5 pp, which is consistent with the narrow gap they report. The tuned-CNN result is best interpreted in the hyperparameter-optimisation literature: Bergstra and Bengio [15] document that random search typically requires budgets in the tens of trials to exceed a strong baseline, and the 8-trial search used here was at the lower end of that range.

7.2 Architectural Comparison

The four architectures cover three design philosophies: a small hand-designed CNN, a short random-search refinement of that CNN, an ImageNet-pretrained EfficientNet with two-stage fine-tuning, and an ImageNet-pretrained DeiT-Tiny transformer fine-tuned end-to-end. On the clean test set the ranking is DeiT (0.9884 / 0.9994) > Custom CNN (0.9837 / 0.9989) > EfficientNet (0.9759 / 0.9974) ≈ Tuned CNN (0.9758 / 0.9974). DeiT and EfficientNet both benefit from ImageNet pretraining, but only DeiT outperforms the Custom CNN, showing that pretraining alone does not determine the ranking. The two lowest-scoring models had partially frozen or under-trained components (only the last 30 EfficientNet layers unfrozen; Tuned CNN search budget

of 8 trials), whereas the Custom CNN and DeiT were trained with all parameters updated and longer schedules.

7.3 Robustness and Scientific Implications

The robustness experiment has practical implications for survey-scale classification. Under noisier labels, both models drop a few accuracy points yet keep AUC above 0.96, so the ranking of probabilities remains useful for downstream review prioritisation. DeiT degrades less than EfficientNet on both metrics (0.6 pp in accuracy; 0.01 in AUC).

The Grad-CAM analysis (Section 6.5) supports the use of EfficientNet as a classifier whose decisions are tied to the target galaxy in both correct cases examined, rather than to background regions, consistent with the case made by Selvaraju et al. [13] for using Grad-CAM as a visual-confidence diagnostic.

7.4 Limitations

Several limitations apply. The study is restricted to a binary classification problem on a balanced subset, which is a simpler task than the full GZ2 decision tree and therefore does not measure performance on multi-class morphology. All experiments use a single random seed (42), so the reported differences between architectures are not accompanied by uncertainty estimates derived from multiple runs. The noise experiment uses only one alternative threshold (0.5) and retrains only two of the four models, because the compute budget on a free Colab T4 was limited. The Grad-CAM analysis reports three qualitative examples; a quantitative pointing-accuracy study would require a systematic comparison of heatmap centroids against catalogue-derived galaxy positions.

7.5 Applications and Future Work

The pipeline generalises to any morphological classification task with labelled imaging. Four extensions address the limitations above: expansion to the full GZ2 decision tree as a multi-label problem [5]; a systematic threshold sweep to yield a robustness curve rather than a two-point comparison; integration of Grad-CAM as a confidence filter for manual review; and application to related tasks such as lens, merger and tidal-feature identification [1][8].

8. Conclusion and Future Work

This project put four deep-learning architectures — a Custom CNN, a Keras-Tuner-optimised CNN, EfficientNetB0 and DeiT-Tiny — through a shared GZ2 pipeline, then re-evaluated the pretrained pair on noisier labels and inspected EfficientNet's decisions using Grad-CAM. The supporting research questions are answered as follows.

RQ1 — Model Accuracy. All four models achieved accuracy above 97% and AUC above 0.997 on the clean test set (Table 1).

RQ2 — Model Comparison. DeiT-Tiny ranked first on both metrics, followed by the Custom CNN. DeiT was fine-tuned end-to-end; EfficientNet had only its top 30 layers unfrozen at stage 2, and the Tuned CNN search was limited to 8 trials.

RQ3 — Robustness. Lowering the GZ2 threshold from 0.8 to 0.5 reduced accuracy by ~7 pp on both pretrained models, yet AUC remained above 0.96 (Table 3); DeiT degraded less than EfficientNet.

RQ4 — Interpretability. Grad-CAM activation concentrated on the target galaxy for both correct predictions, whereas the misclassified example showed diffuse activation over background stars (Figure 11).

Taking the main research question, the findings demonstrate that deep learning classifies binary GZ2 morphologies with accuracy above 97% and AUC above 0.997 on high-confidence labels; DeiT achieves the best scores, and both pretrained models remain informative (AUC > 0.96) under noisier labels.

Future work includes extending to the full GZ2 decision tree, running multi-seed experiments with confidence intervals, sweeping the GZ2 threshold across a continuous range, and integrating Grad-CAM as a confidence filter in an operational pipeline.

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